

Final Project Practice - Pseudocode Submission



Lab Assignment: Team Project Pseudocode

Description:

Your team has been assigned to develop a Java application for a simple library management system. The system should allow users to add, remove, and search for books in the library. Each book has a unique identifier, title, author, and available copies. Additionally, users should be able to check out and return books. The application should store data persistently using a MariaDB database.

Your task for this lab assignment is to submit a pseudocode for the initial design and implementation of the team project. The pseudocode should outline the major steps and logic of the program. Each team member is responsible for contributing to the pseudocode relevant to their assigned tasks.

Pseudocode Requirements:

1. Database Connection (15 points):

- Establish a connection to the MariaDB database.
- Handle database credentials securely.
- Create necessary tables for storing book information.

2. Adding a Book (20 points):

- Prompt the user to enter book details (ID, title, author, copies).
- Validate user input.
- Insert the book information into the database.

3. Removing a Book (15 points):

- Prompt the user to enter the book ID to be removed.
- Validate user input.
- Delete the book information from the database.

4. Searching for a Book (15 points):

- Prompt the user to enter search criteria (title, author).
- Validate user input.
- Retrieve and display matching books from the database.

5. Checking Out a Book (15 points):

- Prompt the user to enter the book ID to be checked out.
- Validate user input.
- Update the database to decrease the available copies.

6. Returning a Book (15 points):

- Prompt the user to enter the book ID to be returned.
- Validate user input.
- Update the database to increase the available copies.

7. User Interface (5 points):

- Design a simple text-based user interface for interacting with the application.
- Display appropriate messages to guide the user.

Submission:

Each team member is required to submit their portion of the pseudocode related to their assigned tasks. The pseudocode should be clear, well-structured, and include comments explaining complex logic.

Grading Rubric (100 points):**• Pseudocode Quality (40 points):**

- Clarity and readability of pseudocode.
- Proper indentation and structure.
- Comments explaining complex logic.

• Database Connection (15 points):

- Proper establishment of a database connection.
- Secure handling of database credentials.
- Creation of necessary tables.
- **Adding a Book (20 points):**
 - Accurate pseudocode for adding a book.
 - Proper user input validation.
 - Database insertion logic.
- **Removing a Book (15 points):**
 - Accurate pseudocode for removing a book.
 - Proper user input validation.
 - Database deletion logic.
- **Searching for a Book (15 points):**
 - Accurate pseudocode for searching for a book.
 - Proper user input validation.
 - Database retrieval and display logic.
- **Checking Out a Book (15 points):**
 - Accurate pseudocode for checking out a book.
 - Proper user input validation.
 - Database update logic.
- **Returning a Book (15 points):**
 - Accurate pseudocode for returning a book.
 - Proper user input validation.
 - Database update logic.
- **User Interface (5 points):**

- Design of a simple text-based user interface.
- Display of appropriate messages.

Total Points: /100

Note: Adjust the rubric based on specific requirements and expectations for the assignment. Provide clear criteria for each section to guide students in meeting the objectives of the assignment.

Points 100

Submitting a file upload

Due	For	Available from	Until
Oct 11	Everyone	-	Oct 22 at 11:59pm
